Fostering Integrated STEM Community as a precursor to a science of politics.

This is delivered in the spirit of a report from the field about a work in progress.

Abstract

Advancing rational inquiry, on MacIntyre's account, requires kinds of communities and fora for their interaction which don't yet exist. There are good reasons for supposing that communities of the right sort could grow at the scale necessary to support the development of consensus around an emerging wholistic account of the human experience. Building and guiding such communities would require the advancement of a political craft beyond any current state of development. In STEM--Science, Technology, Engineering and Mathematics--education there exist precursors to such community and the necessary fora; political craft for building fuller communities can grow there.

I.

MacIntyre has presented an intriguing account of rationality with an esteemed ancient past, a long neglected history and a tantalizing future. Craft-tradition rational inquiry is wholistic, at once practical and theoretical; it progresses through collaborative action within communities whose members together hold and live out substantially common worldviews and aspirations; such a community justifies its claims to have produced the best account thus far by appealing to the history of interactions among mature members from its tradition and from those of its rivals, amassing a store of resources for persuading these mature rivals on their own terms to come eventually to accept its own. Through the establishment of communities of the right sort including civil fora for rival communities to put forward their champions in the *agon* of dialogue among second first-language speakers, the history of rational progress will be written by the native inhabitants and converts of whichever tradition (if any) emerges as thus far most persuasive. The master craft of forming and sustaining any community with claims to rational superiority, which necessarily includes justifying those claims through empirical, quantifiable and persuasive sociological/historical analysis, is the science of politics; the craft of effectively inviting new members into maturity in that master craft is education, and the requisite sort of institution supporting this invitation, the university, on the craft-tradition model.

This craft-tradition account is offered as a lens for viewing both our past and our future. On this account, the prospects for advancing wholistic rational inquiry in our present age are daunting. To put the point bluntly, it can seem that there are no communities and thus no fora for their interaction of the requisite sorts. But advances on several fronts suggest a brighter future for rational inquiry on the craft-tradition model. Here I will briefly sketch three which together set the stage for understanding the timeliness of a fourth, the rebirth of political craft in the context of science education. I will point to a sufficiently robust model of community with at least one actual instance, to advances in collaboration technology that can support the growth of such community, and to advances in instructional design that together point the way toward the near-term construction of a university of a craft-tradition sort, on a seminal scale. The political craft necessary to take this prototypical effort into production can be fostered in a still-robust reflection of fuller human community, through construction at scale of the sort of communities needed to advance science education.

II.

One front is a set of developments that can increase our awareness of the possibilities of long-term voluntary collaborative activity. To convey this awareness I must proceed autobiographically. I have spent the past 32 years on the front of one such development, the emergence of covenant Christian community, in the People of Praise (peopleofpraise.org), itself now more than 38 years in the making. Mature membership in this community involves growing into a lifetime commitment to live out together an emerging worldview and way of life, consisting of as much in common as our commitments to various denominations and churches together with our collective experience enable us to hold and practice in common. Those commitments and that experience support much: we agree together substantially (in freedom and with respect for differences) on matters of theology including principles of interpretation of scripture, cosmology, science and religion, history, psychology and sociology, and morality. The threads holding together this substantial agreement are best discerned through experience with our pattern of life, including weekly meetings, additional small gender-specific group meetings, introductory retreats and subsequent series of some hundred formation workshops, and substantial common participation on economic and service levels. Our agreement is reflected to some degree in the educational philosophy of a school we have founded (trinityschools.org) and articulated in publications we have posted. The metric of our success, acknowledged as the right standard by all, is how well we love one another. Our covenant commitment to one another supports very substantial ties among members; it is not uncommon for members to have dozens of solid friendships that have lasted a quarter century or more. There are several thousand people living this life in some two dozen branches across North America. There are other examples of substantial voluntary associations, intentional communities, but to sketch this first area of development, this single instance will suffice.

Participation in this community life has greatly expanded my sense of the practical limits of voluntary association. But it is not the only set of experiences of this sort. For eight highly formative years far-mostly prior to to my association with the People of Praise, I participated in a tradition of martial arts. I grew up on stories Professors Jiguro Kano and Morihei Ueshiba, and was initiated into a kind of craft with a very substantial view of mature membership, requiring a lifetime of participation in a community of practitioners. (For a visual snapshot pushing in the same direction, find footage of the opening ceremony of the 2008 Olympics in Beijing, once available at http://www.youtube.com/watch?v=8ItRrO6P1W4.) Much later in life these same instincts were reinforced through a 15-year association with a large collaborative science project, the development of the Compact Muon Solonoid (CMS) detector at CERN. That effort involves over 3000 scientists and engineers from 183 institutions in 38 countries, requiring decades of collective investment from design to first collision. Exposure to this and similar large and enduring collaborative research undertakings is therapeutic. It turns out that people can associate with one another however they like; we live in an individualistic age where our sense of what is possible by way of voluntary association is impoverished by our relatively thin experience of community.

The prospects of constructing and maintaining voluntary associations of this and other substantial sorts have been much enhanced as the new web has developed and dramatically lowered the cost of collaboration; this is our a second development. The characteristic mark of the new web is user-generated content. The user/developer distinction has dissolved, so that ordinary users go to the web to post, not just to read. Developers now post functionality as plugable widgets, enabling even the casual user to become a web developer. Ordinary users can now build online collaborations employing the best cutting-edge efficiencies tailored to their own collaborative purposes. And with the integration of voice, texting, imaging, video, rss, search, filtering and other technologies into social networking and collaboration platforms, our capacity for real-time interaction has greatly increased. Every new experience brings users face-to-face with the question, "With whom should I share this?"; and every new task, "With whom should I collaborate?" These pressing questions thrust into relief the possibilities-space for certain dimensions of community life. The new web's potential for community building is largely unexplored, untapped. But that potential is here, now.

A third area of developments concerns assessment-driven project design. The key idea here is "backward design", which means beginning with the end in mind: exactly what do you want to observe happening, and how will you measure that. Project goals are translated into measurable objectives; instruments for measuring the accomplishment of these objectives are designed first, and then program elements are crafted precisely to accomplish program goals operationalized through these objectives as measured by these instruments. Measurement is quantified through use of rubrics that connect qualitative performance descriptions to numerical assessment in ways that can be validated across instruments and for inter-rater reliability. Thus, quantifiable goals drive every step of an assessment-driven project, including the deployment of a curriculum or the building of a community. The more time I spend with backward design people, the more I confuse the best of what they recommend with really clear thinking, and the more it looks like the process scientists observe in large collaborative projects. We are only now just beginning to explore the power of assessment-driven planning for building new substantial forms of voluntary association empowered by new web efficiencies.

These developments suggest a way forward toward the formation of scaleable communities with measurable accounts of mature membership and toward the quantification of their success in engaging rival traditions. I expand on this suggestion in an associated wiki ([[|craft-tradition.wikispaces.com]]) and in the handout; in these few minutes I want to give just a bare sketch. Suppose that people with substantial theoretical agreement live together and put that agreement into practice over a substantial period of time. They may be able to articulate together the main points of their agreement and their experiential basis for accepting the whole system. They could articulate that agreement, and operationalize--translate into measurable goals--the various steps they judge important for entering into their way of life and thought. They could measure the efficiency of their system of invitation to maturity--their education--and hone their curriculum to maximize it. Eventually they could adapt their curriculum to invite mature members from other targeted traditions to climb into 2nd-first-language competency and eventually to join them, accumulating a track record of successes (or failures) in issuing those invitations effectively. At this point the ship has left the dock: quantifiable progress in wholistic rational inquiry would be underway, and the various areas drawn together into the whole system--ethics, religion, politics, everything, included--would be progressing, measurably. None of this seems impossible. It seems to me it could be up and running in a year or two.

Here is a more detailed sketch of what such an environment--community? university? both?--might look like. Imagine an online resource with the following features:
--It is spearheaded by perhaps two dozen people who have lived, worked and worshiped together for 10 years or more and have considered together leading

scholarship in the sciences, history, sociology, psychology, philosophy and theology together, as well as pastoral experience with a community of persons living out the implications of the positions adopted in these areas;

--It contains an overview of a set of positions on what seem to be their most pressing set of questions, perhaps at first glance in a master overview with an associated visual graphic, with supporting area overviews and textual argument available at deeper levels.

--Its construction is collaborative, almost democratized, with a small group of leaders in each area all of whom recognize one another's authority (so far, think wikipedia), in *and among* (now leaving wikipedia) each area of expertise; these say of one another: "if we can't convince you guys, we can't assert this with much confidence."

--Membership as a recognized mature contributor to the conversation is determined by the original group of leaders in each area (recognized by all leaders in every area as competent in each area and in their own) through the establishment of a set of performance outcomes in each area; guidance is provided by a set of milestones with references aiding in the accomplishment of each outcome. (These outcomes are the foundation stones of the edifice in each area; this is backward design--*Understanding by Design*; Wiggins and McTighe.) Outcomes are tied to performance measures through tunable rubrics mapping descriptions of performance onto evaluative categories.

--Performance outcomes in each area come in at least two sets: area competency and area leadership

--New members are admitted as competent contributors in each area, and when they have achieved competency in every area they are admitted as mature members of the community, able to contribute comments which must be taken into account as any area of inquiry advances. The comments of area leaders must be not just addressed but satisfied before the fundamentals of any area of inquiry are changed.

--New areas of inquiry are introduced in a proposal process where comments of any competent members are considered and objections of area leaders are satisfied.

--These requirements for membership and area advancement--call them publication requirements-- can be tuned, perhaps by agreement of key leaders--call them spokespersons--appointed by vote of the leaders of any area of inquiry. Those requirements seem rigid, almost ridiculous, until you compare them with the publication process of large scientific collaborations, against which they seem relatively informal.

--So far, a proposed set of requirements for wholistic rational inquiry inside a community on the craft-tradition model have been identified. The justification for claims they make is internal: their claims satisfy themselves.

--Now suppose that the reference sets for each milestone in each area of inquiry were tuned to address different audiences: language, of course, but also FAQs, textual references, specific arguments and exercises...

--As a first imaginative approximation, imagine this whole system as a contemporary

Summa Theologiae with hyperlinks, textual support, with versions aimed at readers from particular traditions...FAQs tailored to audience, with best practices in assessment embedded.
--Still to come: the requirements of identification of mature membership in rival traditions, and metrics for evidentiary conversion, which requires roughly all of 1) identification as a mature member of a rival tradition (by both the potential antagonist and the leadership of the home tradition in question); 2) advancement to maturity in one's own community; and 3) their registration of a vote--a sustained (?) vote--affirming the superiority of the one system over the other.

There is much hard work to be done, here. But the suggestion is that advancement of all areas of inquiry are open to the prospect of scientific advancement, of quantifiable assessment, taken together as necessary elements of a successful competing whole account, on this picture. Whether these areas actually do advance is a question of how human beings behave under the requisite circumstances. But it now seems practically possible to establish the circumstances necessary to evaluate a tradition's claim to rational superiority as a whole.

IV.

But even if a framework for progress along these lines were well articulated, there would remain a need for a developed craft of attracting participants at the scale needed to implement a full public pilot study and then test for replicability. Rational inquiry seems to progress in the sciences, not in a single laboratory, but in a repeatable, replicable, scaleable way. What are the prospects for rational inquiry on the craft-tradition model moving beyond a nearly private story told among only a small number of mature inhabitants of some happy tradition? Perhaps development on a fourth front will help to brighten the prospects of advancing wholistic rational inquiry at scale.

The scientific community--or a bit more inclusively, STEM Community--is indeed a community with a common good, the advancement of its various interdependent disciplines. STEM community is now famously troubled regarding its care for its own young. In many quarters the community is losing the battle for the hearts and minds of potential new members. (Think of controversies regarding science and religion--on the teaching of evolution, or regarding the evidence for a human contribution to global warming--in the United States.) Ground gained in the development of the craft of initiation of new members--think of the emergence and disappearance of Seaborg's CHEM Study program--becomes lost as the pedagogical community necessary to hand on and advance the craft atrophies. Trouble attracting and retaining teachers, failure to motivate and adequately prepare students as citizens who must serve as the political base for STEM funding, failure to effectively incorporate traditionally uninvited populations, reducing student perception of the craft to one of book learning and hiding almost entirely its advanced practitioners from the bulk of students who could aspire to be like them; the rapid development of eScience and its threat to overwhelm an already substandard STEM educational infrastructure with a new pressing set of demands--this is a partial list of the woes of science and more broadly STEM education in the US.

All of these problems can be viewed as kinds of fragmentation of the community, and the Integrated STEM Community model maintains that the pathway to addressing them is to overcome this fragmentation. Our 15-year experience of a defragmented community--scientists, engineers, K-12 teachers and students working together full time each summer and meeting together weekly year-round--at the Notre Dame QuarkNet Center has produced a new vision of a vibrant STEM community with a rich culture. Among the essential elements of that culture are a research-centered focus; intentionally broad spectra of engagement for professional and lay members with complementary roles in a single community; and a regional professional identity, with complementary professional development tracks for various professional roles. Included among STEM professionals--quite importantly given the current state of the STEM community--are precollege STEM teachers who understand STEM education as the issuing of effective invitations for membership in their own, not someone else's, STEM community. ISTEM culture embraces the full range of collaboration technologies long used to advance the technical disciplines, employing them to craft intentional and integrated STEM community. Finally, given the brevity of this survey, our culture is competitive: the peer review process is a valued sharpening stone against which proposals from all comers are weighed against competitors in a free-for-all of expert review from many and rival perspectives. In this environment, survival has justificatory value. Our last successful proposal, written by K-12 teachers, provided $2.7 million to incorporate graduate students into the ISTEM community at the Notre Dame QuarkNet Center.

V.

ISTEM community has three aspects of significance for the revival of craft-tradition rational inquiry. First, it draws into sharp relief the possibility of substantial community life for many whose backgrounds and mindsets have been characterized by excessive individualism. Secondly, it has the potential for to grow quickly to scale, and is thus a nursery for a precursor of the political craft necessary to build fuller expressions of community life at scale. Finally, with the advent at scale of substantial intentional community built to advance the clear and common good of progress in the STEM disciplines, the limitations of the common good--I say with respect--so narrowly conceived, will become apparent. Hunger for richer dimensions of community life will grow, at scale. And the political craft needed to issue effective invitations into richer intentional communities will have been developing alongside and with assistance from its counterpart in ISTEM community. A renaissance of community life and a modern revival of successful wholistic inquiry on the craft-tradition model will be underway. That's the plan.

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